Ready to build the first real homework session

Open Tutor · product foundation and technical proposal · 3 November 2026

A welcoming tutor. A live conversation. Something useful to show.

Build a small app around LiveKit and reuse selected work we already have.

- Product direction agreed
- Technical choices proposed
- Live integration still to prove



The agreed visual direction: the same companion, without wings, facing the learner.

The next investment should be a working teaching loop, guided by the product and visual foundation.

One URL, then ordinary schoolwork

Windows PC or school Chromebook. No installation or learner account.

01

Arrive

Choose a local learner. The tutor welcomes them.

02

Bring the task

Talk, drop a file, or paste text or a screenshot.

03

Work together

Explain, ask, draw, read, watch, and think quietly.

04

Leave with progress

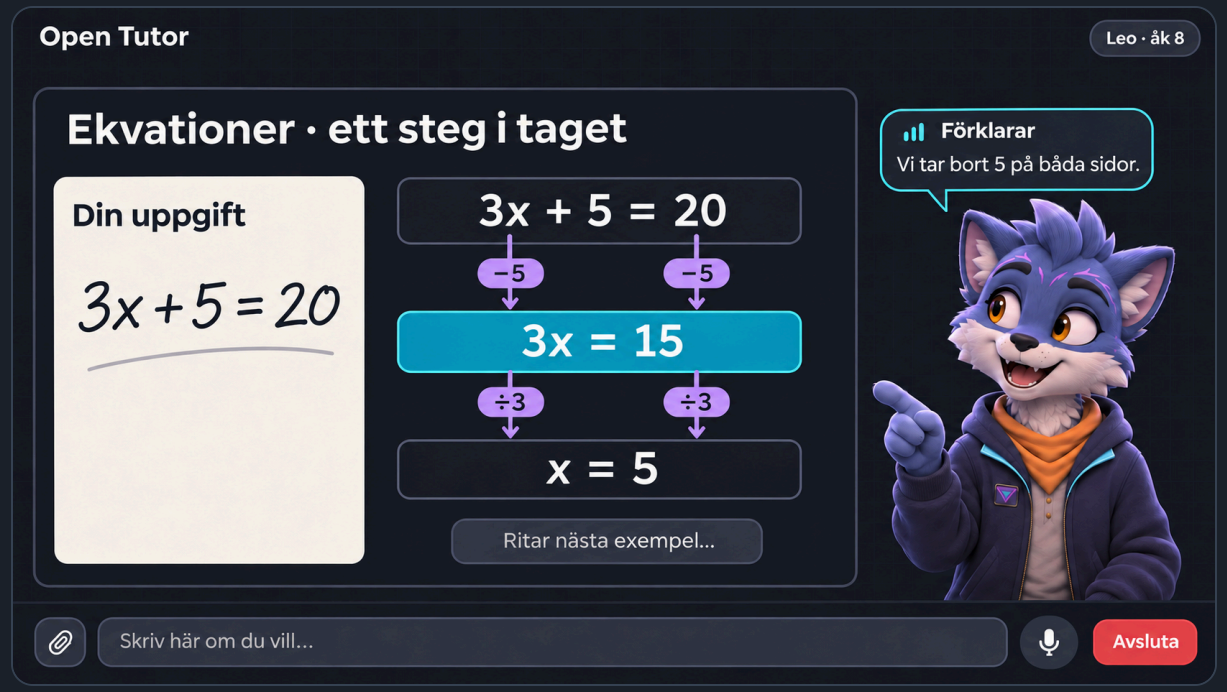
The child does the work. Short learning notes carry forward.

> First use: your child brings whatever is difficult that weekday. No prepared demonstration.

Success means the child understands more and can take the next step themselves.

Keep the companion. Give the lesson room.

The selected second-pass images and style guide are now part of the review package.



Wingless companion on the right, looking into the lesson. Homework and explanation stay readable.

Large on arrival
Welcoming and recognisable.

Smaller during work
The board becomes the main space.

Quiet when useful
No false speaking indicators.

Stable controls
Text, microphone, and end call stay available.

These are design references. Making the interaction work is the next phase.

Build our experience. Borrow selected working parts.

Source inspection points to selective reuse rather than adopting another tutor wholesale.

PRIMARY REUSE SOURCE

Your t3-call

Python voice integrations, session state, visual-stage patterns, and timing instrumentation.

MIT. Leave its T3 and coding workflows behind.

IMPLEMENTATION REFERENCE

AI Math Tutor

Homework context, LiveKit-to-board actions, and explicit turn state.

MIT. Its editable canvas is a different product choice.

IDEAS TO LEARN FROM

Tablo

Multimodal teaching and memory ideas.

AGPL. A larger retrieval stack than we need initially.

Inspected source and licences, not live product sessions. Revisions are recorded in technical evidence.

Reusing patterns saves effort while preserving your specific teaching experience.

Three pieces, with clear responsibilities

The browser shows the lesson. LiveKit carries the call. Our worker coordinates the tutor.

01

Child's browser

Avatar · homework · board
Text and microphone controls
Local learner notes

↔

02

LiveKit Cloud

Live audio
Connection and call state
Board messages and captions

↔

03

Our tutor service

Teaching instructions
Speech and LLM connections
Tools for visuals and resources

Speech recognition · Reasoning model · Speech synthesis · Vision · search · images

The tutor service calls providers. Provider keys stay there.

We can change a voice or model without rebuilding the whole experience.

LiveKit media and data

Familiar parts, each with one job

Choose a small first implementation. Pin exact versions when building it.

The page	React + TypeScript + Vite	Build the approved interface.
The tutor	Python + LiveKit Agents	Manage conversation and tools.
The web service	FastAPI	Serve the page, join calls, receive files.
The board	HTML + SVG + KaTeX	Show precise maths and flexible visuals.
Files and memory	PDF.js + browser storage	Read pages and remember the learner locally.

> Initial speech candidates: Soniox for listening, ElevenLabs for speaking. Choose the actual Swedish voice by listening.

No account platform, central learner database, or local GPU is needed for the first family version.

Vite · FastAPI · PDF.js

A flexible teaching board, without building an editor

The tutor owns the board initially. Children write their assignments where school expects.



The child's own ideas and pasted draft stay distinct from the tutor's explanation.

Fast, precise visuals
Equations, diagrams, steps and mind maps.

Richer content when useful
Generated HTML, animations and images.

Speak in step with the board
Know when a visual is ready before referring to it.

Generated panels stay separate from fixed controls and learner storage. tldraw production licensing is another reason to defer an editor dependency.

Preserve expressive teaching while avoiding an unnecessary canvas-editor dependency.

The LLM needs a supported route for tutoring

The product design survives a provider change. The subscription assumption needs checking.

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OpenCode Go is coding-oriented

Its current documentation describes coding-agent traffic. Tutoring eligibility is unconfirmed.

The Muse Spark Contributor route also lists training use and regional conditions.

Preferred route

Keep the intended Spark candidate if a supported tutoring route is confirmed.

Alternative route

Use an ordinary metered API behind the same provider interface.

OpenCode Go: usage and model terms

Settle the route before real homework. No provider settings or credentials have been changed.

Use the home lab, with managed call transport

A proposed deployment, not a service that is live today.



The page and tutor service

Proposed URL: **tutor.labtopia.dev**

Use CPU hosting already available. Confirm machine availability and DNS during deployment.

The live call

Browser audio connects through LiveKit Cloud directly.

Inference runs with providers; no GPU installation on the child's PC.

> Infisical access is confirmed. Speech-provider key names are present. Key validity, balances, and LiveKit provisioning still need checking.

Start with family use. A public demo can follow with a simple owner-controlled spending limit.

Four milestones, each ending in something usable

This package completes the foundation and technical proposal. Implementation starts after review.

A

Prove the loop

Swedish voice, homework page, interruption, quiet work and a board update together.

NEXT

B

Complete the workspace

Approved character, profiles, uploads, teaching board, links and learning notes.

C

Put it behind a URL

Persistent hosting and checks on the actual Windows PC and Chromebook.

D

Use real homework

Observe a weekday session and fix the friction the child actually encounters.

Milestones describe results, not promised dates. No integration experiment has been run yet.

The first working session is the priority; elaborate animation and expansion come later.

A grounded proposal, with the unknowns visible

We have inspected the ingredients. We have not proved the recipe.

KNOWN

Inspected

- PRD and chosen visual direction
- Source and licences in three repositories
- Current integration documentation
- Infisical project access and key names

PROPOSED

Recommended

- Small app with selected reuse
- Python worker and React page
- Tutor-controlled web board
- Home-lab hosting first

TO PROVE

Next experiment

- LLM access and streaming
- Swedish voice quality
- Quiet work and interruption
- Homework vision and visual timing

There is no measured cost, latency claim, deployed app, or live teaching result in this package.

A concrete direction to take into the first build

The product ambition stays intact. The implementation starts small.

01 **Build the small app, borrowing selected parts**

Preserve the experience you chose instead of inheriting another tutor's whole product.

02 **Resolve the model route, then test the loop**

Keep providers replaceable and choose the Swedish voice through actual listening.

03 **Host the family version on the existing lab**

Reach a normal browser URL; consider wider access after the family experience works.

Review PR #1 Technical plan PRD Image gallery

Next authorised build phase, after your review: the integration experiment. Nothing has been deployed.